BRAC University
Department of Computer Science and Engineering

Semester: **Spring 2023**
Course Code: **CSE460**
Course Title: **VLSI Design**

Midterm Exam
Full Marks: **15 x 3 = 45**
Time: **1 hour 30 minutes**
Date: **4th March 2023**

Set A

Student ID:	Name:	Section:

[Answer any **THREE** questions out of **FOUR**. Each question carries equal marks.]

[After the exam, the question paper should be turned in along with the answer script.]

1. (CO1) The timing diagram of an edge triggered sequential circuit element is shown in **Figure 1**

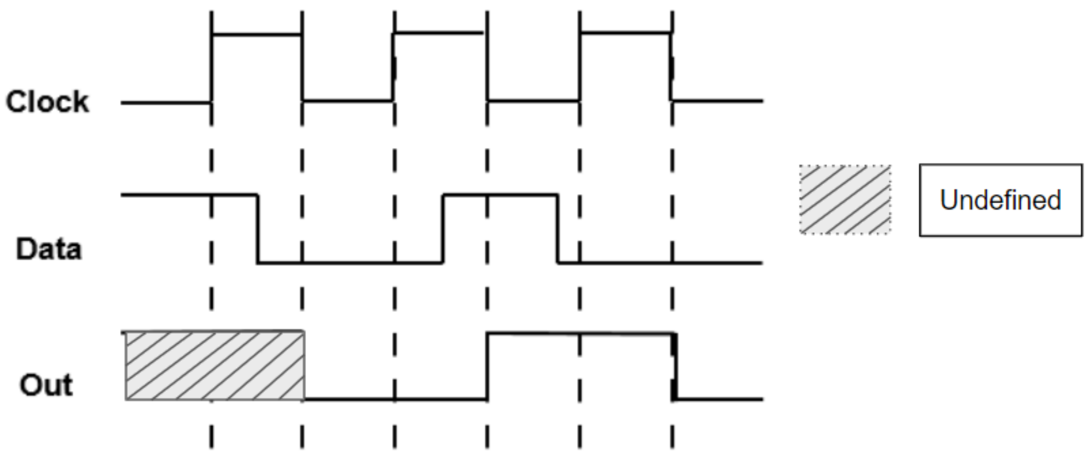

Figure 1

(a)	**Identify** the type of the edge triggered circuit element shown in **Figure 1**.	**3**
(b)	**Justify** if it is possible to build the circuit element as shown in **Figure 1** using level triggered sequential circuit elements only. **Draw** the appropriate block diagrams.	**6**
(c)	**Draw** the appropriate circuit diagrams corresponding to the block diagrams of **(b)** using transmission gates.	**6**

2. (CO3) Observe **Figure 2** and answer the following questions.

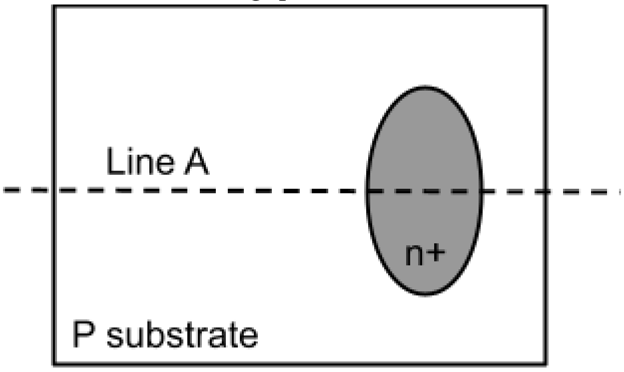

Figure 2: Top view of a wafer after fabricating an n+ diffusion region (dark region)

2. (CO3)	(a)	**Draw** the cross-sectional view of the above figure along 'Line A' and **label** the different regions.	3
	(b)	Briefly **describe** and **illustrate** the necessary steps to fabricate the n+ diffusion region on a blank wafer as shown in the figure.	12

3. (CO3)

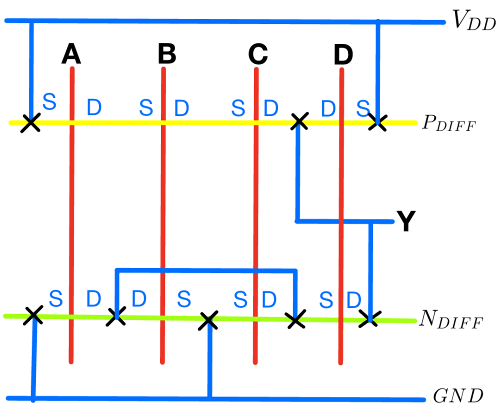

Figure 3

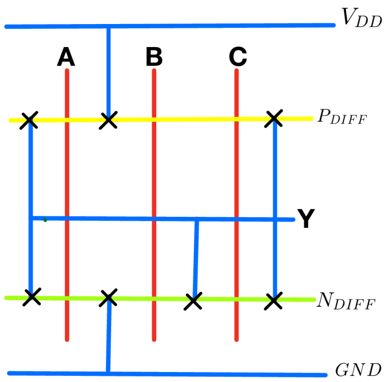

Figure 4

3. (CO3)	(a)	i) **Design** the CMOS compound gate for output Y from the given stick diagram in **Figure 3**. Clearly indicate the PUN and PDN networks.	4+4
		ii) **Find** the layout area of **Figure 3** in terms of Lambda (λ).	1
	(b)	**Identify** the connection **errors** in the stick diagram for the compound gate, **Y**= $\overline{AB + C}$ in **Figure 4**. Design the **correct** stick diagram for **Y**.	6

<table>
<tr><td>4. (CO2)</td><td colspan="2">You are tasked with designing a CD/DVD drive's control circuitry using a Moore-type finite-state machine. The drive has one input push button w, which generates $w = 1$ when pushed, and $w = 0$ when un-pushed. A $w = 1$ input causes the tray to open if it was already closed, or causes the tray to close if it was already open. The control circuit has only one output signal, z, which determines if the CD/DVD drive should be open ($z = 1$) or closed ($z = 0$) with the following instructions:
● If the tray is closed, pushing the button (setting $w = 1$) should generate the signal ($z = 1$) to open the tray
● If the tray has been open for less than 3 clock cycles, pushing the button (setting $w = 1$) should generate the signal ($z = 0$) to close the tray
● The tray should automatically close after being open for 3 clock cycles
● For simplicity, assume that the time required for opening and closing the tray is negligible compared to the system clock's time period

</td><td></td></tr>
<tr><td></td><td>(a)</td><td>Determine the frequency of the system clock if the tray won't stay open for more than 5 seconds.</td><td>2</td></tr>
<tr><td></td><td>(b)</td><td>Draw the state diagram for the machine.</td><td>5</td></tr>
<tr><td></td><td>(c)</td><td>Derive a state-assigned table from (b) using the gray-encoding technique.</td><td>3</td></tr>
<tr><td></td><td>(c)</td><td>Find the next-state and output logic expressions using k-map.</td><td>5</td></tr>
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